

# Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration  
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 11

|                     |                   |
|---------------------|-------------------|
| Student ID          | IT24104150        |
| Student Name        | Sathsaranee K.L.U |
| Campus/ Center name | Malabe            |
| Specialization      | CSNE              |
| Batch No            | Y2S2.WD.CSNE.01   |

## Activity 1

```
vboxuser@ubuntu: ~
GNU nano 7.2 activity1.sh
#!/bin/bash
while true
do
    echo "----- MENU -----"
    echo "1. Show Today's Date/Time"
    echo "2.Show files in current directory"
    echo "3.Show calendar"
    echo "4.Start editor to write letters"
    echo "5.Exit"
    echo "Enter your choice: "

    read choice

    case $choice in
        1)
            date
            ;;
        2)
            ls
            ;;
        3)
            cal
            ;;
        4)
            nano letter.txt
            ;;
        5)
            echo "Exiting..."
    esac
done
```

```
vboxuser@ubuntu:~$ nano activity1.sh
vboxuser@ubuntu:~$ chmod +x activity1.sh
vboxuser@ubuntu:~$ ./activity1.sh
----- MENU -----
1. Show Today's Date/Time
2.Show files in current directory
3.Show calendar
4.Start editor to write letters
5.Exit
Enter your choice:
```

## Activity 2

### patteren 1

```
vboxuser@ubuntu: ~  
GNU nano 7.2 patteren1.sh  
#!/bin/bash  
for ((i=1;i<=5;i++))  
do  
  for ((j=1;j<=i;j++))  
  do  
    echo -n "$i"  
  done  
  echo  
done
```

```
vboxuser@ubuntu:~$ nano patteren1.sh  
vboxuser@ubuntu:~$ chmod +x pattern1.sh  
vboxuser@ubuntu:~$ ./pattern1.sh  
1  
22  
333  
4444  
55555
```

### Pattern 2

```
GNU nano 7.2 patteren2.sh  
#!/bin/bash  
for ((i=1;i<=5;i++))  
do  
  for ((j=1;j<=i;j++))  
  do  
    echo -n "$j"  
  done  
  echo  
done
```

```
vboxuser@ubuntu:~$ nano patteren2.sh  
vboxuser@ubuntu:~$ chmod +x patteren2.sh  
vboxuser@ubuntu:~$ ./patteren2.sh  
1  
12  
123  
1234  
12345
```

### Pattern 3

```
GNU nano 7.2                                pattern3.sh
#!/bin/bash
for ((i=5;i>=1;i--))
do
  for ((j=1;j<=i;j++))
  do
    echo -n "| "
  done
  echo
done
```

```
vboxuser@ubuntu:~$ nano pattern3.sh
vboxuser@ubuntu:~$ chmod +x pattern3.sh
vboxuser@ubuntu:~$ ./pattern3.sh
|||||
|||| 
|||  
||   
||  
|
```

### Pattern 4

```
GNU nano 7.2                                pattern4.sh
#!/bin/bash
for ((i=1;i<=5;i++))
do
  for ((j=1;j<=i;j++))
  do
    echo -n "*"
  done
  echo
done
for ((i=4;i>=1;i--))
do
  for ((j=1;j<=i;j++))
  do
    echo -n "*"
  done
  echo
done
```

```

vboxuser@ubuntu:~$ nano pattern4.sh
vboxuser@ubuntu:~$ chmod +x pattern4.sh
vboxuser@ubuntu:~$ ./pattern4.sh
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

### Activity 3

```

GNU nano 7.2 activity3.sh
#!/bin/bash
if [ $# -lt 3 ]
then
    echo "Error: Please provide three numbers"
    exit
fi
a=$1
b=$2
c=$3

largest=$a

if [ $b -gt $largest ]
then
    largest=$b
fi

if [ $c -gt $largest ]
then
    largest=$c

```

```

vboxuser@ubuntu:~$ nano pattern4.sh
vboxuser@ubuntu:~$ nano activity3.sh
vboxuser@ubuntu:~$ chmod +x activity3.sh
vboxuser@ubuntu:~$ ./activity3.sh 10 25 15
Largest number is: 25

```

## Activity 4

```
vboxuser@ubuntu: ~
GNU nano 7.2 activity4.sh
#!/bin/bash
show_info() {
    echo "Day and Time:"
    date

    echo "Login User:"
    whoami

    echo "Current Directory:"
    pwd
}
show_info
```

```
vboxuser@ubuntu:~$ nano activity4.sh
vboxuser@ubuntu:~$ chmod +x activity4.sh
vboxuser@ubuntu:~$ ./activity4.sh
Day and Time:
Tue Apr 21 09:53:10 AM UTC 2026
Login User:
vboxuser
Current Directory:
/home/vboxuser
```